



Healthcare Data Science Report

Assignment 1 – Problem Definition

Stroke Risk Stratification and Prevention Strategies Based on Multidimensional Features ——Decision Support for Stroke Prevention in Healthcare, Industry, and Society

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1. Motivation / Importance of the Data

According to the Global Burden of Disease (GBD) Study 2019 [1], the top four causes of death are ischemic heart disease (13.7%), stroke (10.8%), chronic obstructive pulmonary disease (6.1%), and lower respiratory tract infections (5.6%). Figure 1.1 indicates that the incidence of stroke is remarkably high. In 2021, stroke remains the second leading cause of death (about 7 million) and the third leading cause of death and disability combined (as expressed by disability-adjusted life-years lost—DALYs; over 160 million DALYs) in the world [2] [3]. Stroke results from acute vascular damage to the central nervous system (e.g., cerebral infarction, intracerebral hemorrhage) [4], leading to complications such as falls, infections [5], and limb movement difficulties (affecting 80% of survivors), severely impacting patients' quality of life and daily functional abilities [6].

The economic burden of post-stroke care is substantial—approximately 34% of global healthcare expenditure is spent on stroke treatment, with costs in the U.S. reaching up to \$140,000 per stroke incident (including hospitalization and rehabilitation) [7]. Stroke incidence is expected to rise from 2020 to 2030, highlighting the inadequacy of current primary prevention strategies and the urgent need to strengthen global preventive measures [8]. The study also showed that stroke is highly preventable, with 84% of the stroke burden in 2021 attributable to 23 modifiable risk factors, including high blood pressure, air pollution, and excess body weight—presenting a public health challenge and an opportunity for action [3].

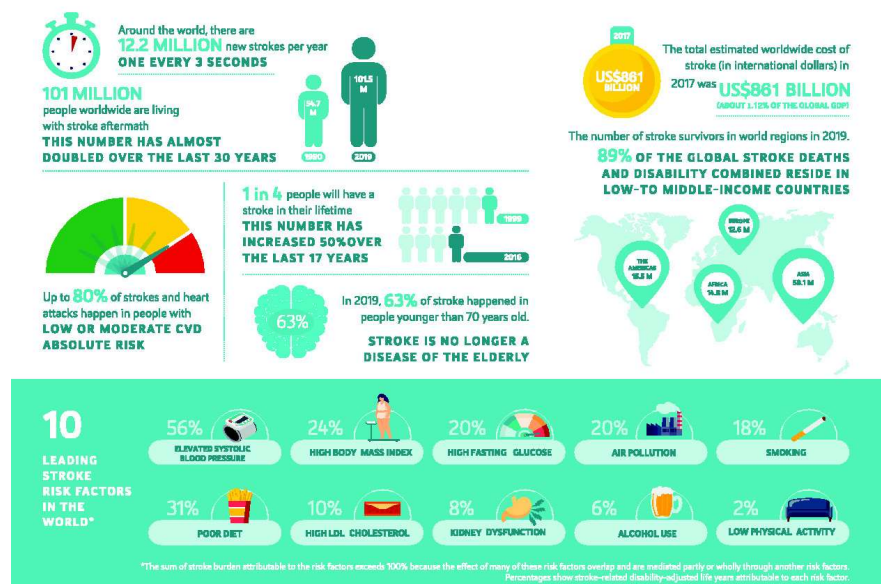


Figure 1.1: World Stroke Organization (WSO):
Global Stroke Fact Sheet 2022 [3]

With the development of medical technology and data science, machine learning and artificial intelligence (AI) have great potential in stroke prediction. Using stroke datasets, we can develop prediction models based on multiple features to identify high-risk groups and provide disease prevention, early intervention, personalized medical support, optimize medical resources, and provide a basis for scientific research and policy-making.

Based on this, the stroke dataset can be regarded as an important information resource for government and non-government stakeholders to support decision-making and medical resource allocation, such as stroke prevention measures, personalized health management programs, and early screening projects tailored to the needs of specific populations



2. Research Question

This study focused on optimizing stroke risk assessment and prevention strategies. It aimed to construct an accurate hierarchical stroke risk model based on multidimensional characteristics. Data from Kaggle's Brain Stroke Dataset [9]. The goal was to provide a scientific basis for early warning, precise prevention and control of stroke, and the formulation of public health policies. Based on that, we will use four sub-questions to answer the research questions.

The results of these studies can provide clearer data portraits of people at higher risk of stroke and help stakeholders decide what to do next faster. The sub-questions all wanted to find out which factors would have a stronger effect on stroke occurrence.

Sub-question 1: The Relationship between Chronic Underlying Diseases and Stroke Risk - Rui Zhang

This sub-question study focuses on the chronic underlying diseases (hypertension, heart disease, and glucose level in blood) and systematically explores their association mechanisms with brain stroke risk. Firstly, we will explore the impacts and associations of the above chronic diseases respectively with brain stroke risk. Additionally, we will analyze data to determine the degree of correlation between different diseases and stroke individually, ascertain whether there are direct or indirect effects, and quantify the magnitude of these effects.

Furthermore, we aim to develop a multi-factors model to analyze the combined associations that contribute to brain stroke risk. It is important to reveal the complex interactions among these factors and investigate efficient feature engineering work to enhance the prediction of brain stroke risk.

Sub-question 2: The Relationship between Lifestyle Factors and Stroke Risk - Zhuoxiao Chen

This research sub-question will focus on two core lifestyle factors—smoking status and BMI—systematically investigating their association with stroke risk. Initially, it precisely identifies the differential impacts of various smoking behavior patterns on stroke risk; subsequently, it employs BMI stratification criteria to elucidate the dose–effect relationship between body mass index and stroke risk. Furthermore, by incorporating an interaction model, the study primarily examines the joint exposure effects of smoking and BMI. Through stratified analyses comparing risk increments among smokers across different BMI levels, it clarifies the synergistic mechanism by which these factors collectively amplify stroke risk.

Sub-question 3: Health Equity Analysis of Socio-economic Factors and Brain Stroke Risk - Harriet Luo

In this study, we will use *work_type*, *Residence_type*, and *ever_married* as key factors to develop a simple composite indicator of socio-economic status. We will systematically analyze these social structural factors to investigate if there are differences in brain stroke risk between social groups and assess the presence of health inequity.



We will examine the relationship between socio-economic factors and brain stroke risk in different age and gender subgroups, use this dataset to analyze how these factors affect stroke risk. The expected findings will help measure the effect of socio-economic status and show patterns of health inequity.

Sub-question 4: What role does age play in diagnosing stroke patients? - Jiayu Xiong

In this subproblem, we focus on age and explore whether there are significant differences in stroke occurrence rates among different age groups (e.g., 0-30, 31-50, 51-70, 71+). Data analysis will look for an age threshold at which stroke risk increases significantly after this, as key evidence to alert stakeholders.

In addition, whether there is interaction between age and other factors is also one of the objectives of the question. It is also important to pay attention to the relationship between high age and other physical diseases (such as higher blood sugar levels, high blood pressure, etc.) and whether there is a correlation with stroke.

3. Stakeholders

In view of stroke being a major global chronic disease threat, its prevention and control system has transcended a single medical dimension to establish a full-lifecycle and collaboratively governed health system encompassing “prevention–screening–treatment–rehabilitation–management”. The system is built upon patient data as its digital cornerstone. It has gradually achieved the leap from population-based prevention to individualized precision medicine, thereby inaugurating a new era of data-driven advances in biomedical industrial innovation. Below are the key stakeholders at various stages of the stroke management system:

Prevention and Health Education: Government and Public Health Institutions, Primary Hospitals

In this stage, the stakeholders could leverage patient data to create community stroke risk heatmaps and enable targeted interventions. The policy evaluation could be executed by integrating multidimensional data to assess the impact of policies on stroke incidence among specific populations.

Acute Phase Treatment: Multidisciplinary Teams in Hospital

In acute treatment, teams could develop stroke recurrence risk scoring models to refine medication strategies, and utilize gender and age factors to optimize pre-screening processes in emergency departments, thereby enhancing treatment efficiency.

Rehabilitation and Digital Long-Term Care: Rehabilitation Hospitals, Home Care Platforms and Digital Therapy Enterprises

Hospitals formulate comprehensive rehabilitation plans by integrating data from multiple etiological factors. In conjunction with the data, the core algorithm for a cognitive training application can be developed to deliver personalized dietary and exercise plans to individual patients [10]. By analyzing

the data, intervention strategies are tailored to distinct user profiles. Rehabilitation recommendations could be dynamically adjusted real-time data to improve personalized remote consultation support.

Pharmaceutical Research and Innovation: Research Institutes, Universities, Pharmaceutical Companies, Medical Device Companies

Researchers can leverage data to investigate the pathogenic mechanisms in greater depth. Companies can utilize these insights to design targeted therapeutic drugs and intraoperative materials. Subsequently, they can formulate differentiated market promotion strategies tailored to diverse etiologies, thereby enhancing product coverage and clinical applicability.

Policy and Insurance Support: Governments' Medical Insurance Groups, Insurance Companies

These stakeholders could formulate flexible insurance plans and adjust premium discounts based on individual health data.

Patient Support: Patients and Their Families

Patients and their families can utilize data to dynamically assess recurrence risk and refine preventive strategies. Subsequently, they can develop personalized dietary regimens and home-based exercise programs to support long-term health management and improve overall well-being.

4. Project Pipeline

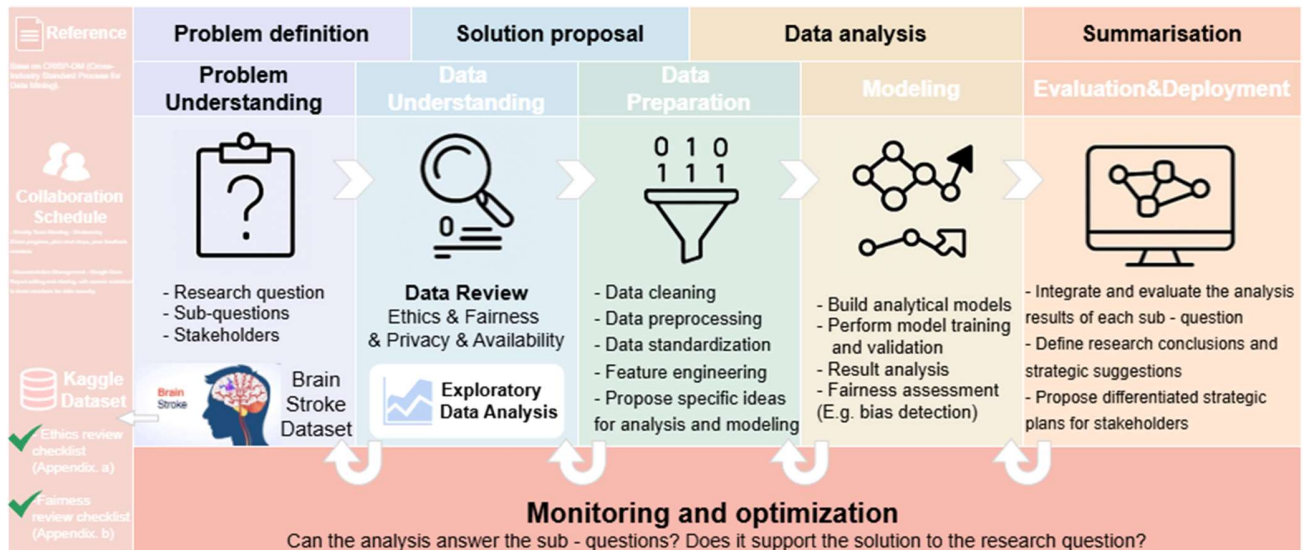


Figure 4.1 pipeline

Figure 4.1 shows the data science pipeline used in our project, following the CRISP-DM method [11]. As a widely used framework in data mining, CRISP-DM highlights the importance of repeating feedback across key stages: understanding the data, preparing it, building models, evaluating



results, and finally deploying them. This cycle-based structure has been found to boost the success rate of data science projects [12].

This project aims to find stroke-prone groups by analyzing patients' diverse features and to create tailored prevention strategies for various socio-economic segments. The process covers everything—from problem definition to strategy development—while we remain mindful of privacy, ethics, and fairness at every step.

We use a public stroke dataset from Kaggle, this dataset was provided by community contributors. It contains typical clinical variables and was initially processed before its public release. Before conducting the analysis, we reviewed the dataset with a focus on ethics, privacy, and fairness (see Appendix a & b for the checklist). These concerns are especially critical during the data preparation stage, where steps like de-identification and anonymization can greatly affect how fair and private the models are [13]. In later phases, we continue to enforce privacy and fairness, staying compliant with data laws and avoiding built-in bias [14].

Effective management and coordination are often what make or break data science projects [12]. To address this, we built a feedback loop that supports flexible changes at any stage. This system not only boosts management efficiency but also helps each phase achieve its purpose, reducing the risk of miscommunication or failure across steps [14], [12].

5. Expected Outcomes

Since the data set does not track the information collected by the respondents and was last updated three years ago, we based our correlation analysis on a cross-sectional time node, so the conclusions drawn may not be fully consistent with the current situation. We will explore in depth the influence of each attribute on the probability of having a stroke, and analyze from multiple dimensions which groups of people are more likely to have a stroke. For example, from the perspective of social economy, personal physical health, lifestyle, and age.

Based on these different perspectives, we can get a portrait of people who are at risk of stroke and people who are not at risk of stroke, and provide the portrait of the population to stakeholders. Whether it is preventive advocacy for people with a low risk of stroke or medical diagnosis for people with a high risk of stroke, stakeholders can rely on the visualizations produced in this report to make more effective actions and judgments.

6. References List

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7. Appendix

a. Ethics and Privacy Checklist

Ethics and Privacy Checklist		
No.	Review Item	Compliance
1	Data source is clear, credible, and properly cited.	✓ Compliant
2	Dataset has a clear usage license (Database Contents License (DbCL) v1.0).	✓ Compliant
3	Data collection purpose is clear and appropriate for medical research.	✓ Compliant
4	Dataset contains no directly identifiable personal information.	✓ Compliant
5	Sensitive information (age, gender, BMI, chronic disease history) has been appropriately de-identified.	✓ Compliant
6	Dataset has no apparent risk of re-identification, or measures have been taken to minimize this risk.	✓ Compliant Avoid conducting any data analysis that may increase the risk of re - identification or disclosing details of the public data.
7	Data storage and usage comply with information security and privacy protection requirements.	✓ Compliant
8	IRB approval	⚠ None but this usage is limited to the purpose of in - class learning only.

b. Fairness Checklist

Fairness Checklist			
No.	Review Item	Result	Notes
1	Gender distribution balance	✓ Female: 58.4%, Male: 41.6%	Slight imbalance, but acceptable
2	Residence (Urban/Rural) distribution balance	✓ Urban: 50.8%, Rural: 49.2%	Very balanced
3	Work type distribution balance	⚠ Dominated by "Private": 57.4%	Need awareness in analysis; could introduce bias
4	Smoking status distribution balance	⚠ "Unknown" category present	Consider excluding or addressing "Unknown" status
5	Missing data patterns	⚠ "Unknown" smoking status	Review missing data patterns; possibly impute
6	Overall fairness & representativeness	✓ Acceptable	Regular monitoring recommended for sensitive attributes